

1.Aim

To demonstrate Locard's Exchange Principle by displaying the metadata (created, modified, and accessed timestamps) of a file, showing that every interaction with a file leaves digital traces.

Algorithm

1. Start the program.
2. Create or open a file.
3. Write data into the file.
4. Read the file metadata using the operating system.
5. Display the file creation time.
6. Display the file modification time.
7. Display the file access time.
8. Stop the program.

Code

```
import os
```

```
import time
```

```
file_path = "sample.txt"
```

```
# Create a file
```

```
with open(file_path, "w") as f:
```

```
    f.write("Digital Forensics Demo")
```

```
# Get file metadata
```

```
stats = os.stat(file_path)
```

```
print("File Metadata (Trace Evidence):")
```

```
print("Created Time :", time.ctime(stats.st_ctime))
```

```
print("Modified Time:", time.ctime(stats.st_mtime))
```

```
print("Accessed Time:", time.ctime(stats.st_atime))
```

Output

File Metadata (Trace Evidence):

Created Time : Tue Jul 21 08:01:48 2026

Modified Time: Tue Jul 21 08:01:48 2026

Accessed Time: Tue Jul 21 08:01:48 2026